Load Dynamics: Minimal Equations and Stability Conditions in Dimensional Human Field Theory (DHFT)

DHFT Technical Note: Minimal Equations & Stability Conditions

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Abstract

This document defines the minimal equations and stability conditions governing system behavior in Dimensional Human Field Theory (DHFT). The system is specified over invariant variables Load (L), Capacity (C), and Boundary (B).

The equations presented here formalize the conditions under which a system remains stable, enters instability, or undergoes collapse. These conditions are expressed in terms of load–capacity relations, boundary-regulated flow, and temporal accumulation.

No additional variables are introduced. All equations remain constrained by the minimal system defined in DHFT.

I. Scope

This document addresses only the minimal conditions required to determine whether a human system remains stable, enters instability, or undergoes collapse under load. It does not address meaning, interpretation, identity content, moral evaluation, or domain-specific intervention.

The model functions as a first-order anchor. It specifies the minimum invariant structure necessary to describe system behavior under demand while preserving non-collapse constraints and avoiding explanatory inflation.

II. Variable Reference

The system is defined over three invariant variables: Load (L), Capacity (C), and Boundary (B). These variables are treated as structurally independent but causally related.

III. Stability Conditions

System stability, instability, and collapse are determined by the relation between load and capacity under boundary conditions.

The stability conditions are defined as follows:

$$L \leq C \tag{1}$$

$$L > C \tag{2}$$

$$L > C \text{ over interval } t \tag{3}$$

IV. Boundary-Regulated Flow

Load accumulation within a system is determined by the relation between inflow and outflow under boundary conditions.

Net load is expressed as follows:

$$L\_\text{net} = L\_\text{in}(B\_\text{in}) - L\_\text{out}(B\_\text{out}) \tag{4}$$

Net accumulation occurs when inflow exceeds outflow under boundary conditions.

V. Rate Condition

System behavior depends not only on load magnitude but also on the rate of load accumulation.

The rate of change of load is defined as follows:

$$\mathrm{d}L/\mathrm{d}t \tag{5}$$

Instability may arise when the rate of load accumulation exceeds the system's capacity to process it.

VI. Temporal Accumulation

System state depends on cumulative load over time rather than instantaneous load alone.

Cumulative load is expressed as follows:

$$\int \text{ from } t_0 \text{ to } t_{\square}\; L(t)\, \mathrm{d}t \tag{6}$$

Accumulated load over an interval contributes to instability even when instantaneous load remains within capacity.

VII. Constraint

All equations presented reduce to Load (L), Capacity (C), and Boundary (B). No additional variables are introduced.

The system remains closed under these relations and does not permit expansion of the variable set at the minimal layer.

VIII. Statement

The equations defined here provide a minimal and sufficient formalization of system dynamics in DHFT. All valid system descriptions at the minimal layer reduce to these relations. All valid descriptions of system behavior at the minimal layer reduce to these relations.

IX. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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